



INTERNSHIP PRESENTATION

Machine Learning Applications in Intelligent Toll Management Systems

KENT Intelligent Transportation Systems (India) Pvt. Ltd.

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Internship Overview

Machine Learning track applied to an Intelligent Transportation Systems context

01 Organization

KENT Intelligent Transportation Systems (ITS)

02 Domain

Toll automation, electronic toll collection, AV

03 Core Work

Data preparation, EDA, classification, predic

04 Outcome

Practical ML workflow, evaluation and docum

Objective

Build practical Python/data-analysis skills and understand how ML methods can support intelligent toll and transportation workflows.

KENT & Intelligent Toll Automation

Domain understanding used to frame the internship work

A

Toll Automation

Vehicle detection and classification, lane control, transaction

B

Toll Xplore

Toll collection platform described in the report with transaction

C

Traffic Management

Real-time road, traffic and weather information using counters, cameras

Vehicle / Sensor Data

AVC / ETC / Lane Systems

Data & Analytics

ML Support

Operational Review

Machine Learning Workflow

A compact end-to-end methodology followed across the internship tasks

Problem Definition

Data Collection

Cleaning

EDA

Feature Engineering

Train / Test

Model Training

Evaluation

ML

Classification

Vehicle class as the target. Decision Tree and Random Forest

ML

Regression

Traffic volume as a numerical target. Linear Regression pro

ML

Anomaly Detection

Isolation Forest identifies observations that differ from expected

Technology stack: Python • NumPy • Pandas • Matplotlib • Scikit-learn • Jupyter / VS Code • Git

Data Preparation & Exploratory Analysis

Clean the data first, then understand patterns before modelling

01 Data Features

Vehicle_Length
Vehicle_Height
Axle_Count
Lane_ID
Vehicle_Class

Hour • Traffic_Volume • Payment_Mode • Day/time context

02 Preprocessing

Missing-value handling
Duplicate / invalid-record checks
One-hot encoding
Scaling where required
Avoiding data leakage

03 EDA

Vehicle-category distribution
Traffic volume by hour
Numerical summaries
Correlation analysis
Patterns, imbalance and unusual observations

Core Machine Learning Applications

Two practical supervised-learning problems in the internship track

01 Vehicle Classification

INPUTS

Length + Height + Axles

MODEL

Decision Tree / Random Forest

OUTPUT

Predicted vehicle class

EVALUATION

Accuracy • Precision • Recall • F1-score • Confusion Matrix

02 Traffic Volume Prediction

INPUTS

Hour • Day • Lane • Historical Volume • Holiday • Weather

MODEL

Linear Regression baseline

OUTPUT

Estimated traffic volume

METRICS

MAE • RMSE • R^2

Anomaly Detection & Model Evaluation

Detect unusual observations and measure model quality systematically

A

Anomaly Detection — Isolation Forest

Examples:

- Unexpected traffic volume
- Unusual transaction timing
- Sensor measurements outside normal ranges

Important: an anomaly is a review signal — not proof of fraud, wrongdoing or a confirmed threat

E

Evaluation Framework

CLASSIFICATION

Accuracy • Precision • Recall • F1-score • Confusion Matrix

REGRESSION

MAE • MSE • RMSE • R^2

Model comparison considers both performance and interpretability.

Integrated ML Architecture

Conceptual connection between transportation data and ML-supported decisions

Vehicle / Sensor Data

Acquisition

Validation

Preprocessing

Features

ML Model

Output

Dashboard / Review

01

Implementation

Separate X and y; use train/test split; apply transformations

02

Human Oversight

ML outputs should support operational decisions and be mo

03

Future Direction

Computer vision, Edge AI, Explainable AI, time-series/deep lea

Learning Outcomes & Demonstration Evidence

Selected evidence is based on synthetic DEMO DATASET examples

01 Technical Growth

Python, tabular data, preprocessing, EDA, classification, regression

02 Professional Growth

Documentation, debugging, problem decomposition, reproducible

03 Future Scope

Computer vision, edge AI, explainability, time-series models and



Internship Certificate & Conclusion

Completion certificate and final takeaway



Internship Achievement

The internship provided a structured foundation in Machine Learning while connecting data-driven me

Key Takeaway

Understand the problem → prepare data → explore → model → evaluate → interpret → document.

THANK YOU

Questions & Discussion